Case Study 3: The Conjugate Gradient Method

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Link to Github Repository:

https://github.com/mclane2/Case-Study-3-The-Conjugate-Gradient-Method

Question 1

I implemented the Conjugate gradient algorithm from section 6.7 of Saad's textbook in the .py file "CG_Algorithm_function.py". It takes the matrix A, the right-hand side vector b, the tolerance (`tol`), and the maximum number of iterations (`max_iters`) as arguments.

Question 2

Figure 1 shows the convergence of the conjugate gradient algorithm for matrices A of size N = 512, 1024 and 2048. The relative residual norm was calculated as magnitude of the residual r divided by the magnitude of the RHS vector b, namely

$$\text{relative residual norm} = \frac{||b - Ax_i||}{||b||}$$

and saved every iteration. Figure 1 shows that the smaller sized matrices converge faster then the larger matrices.

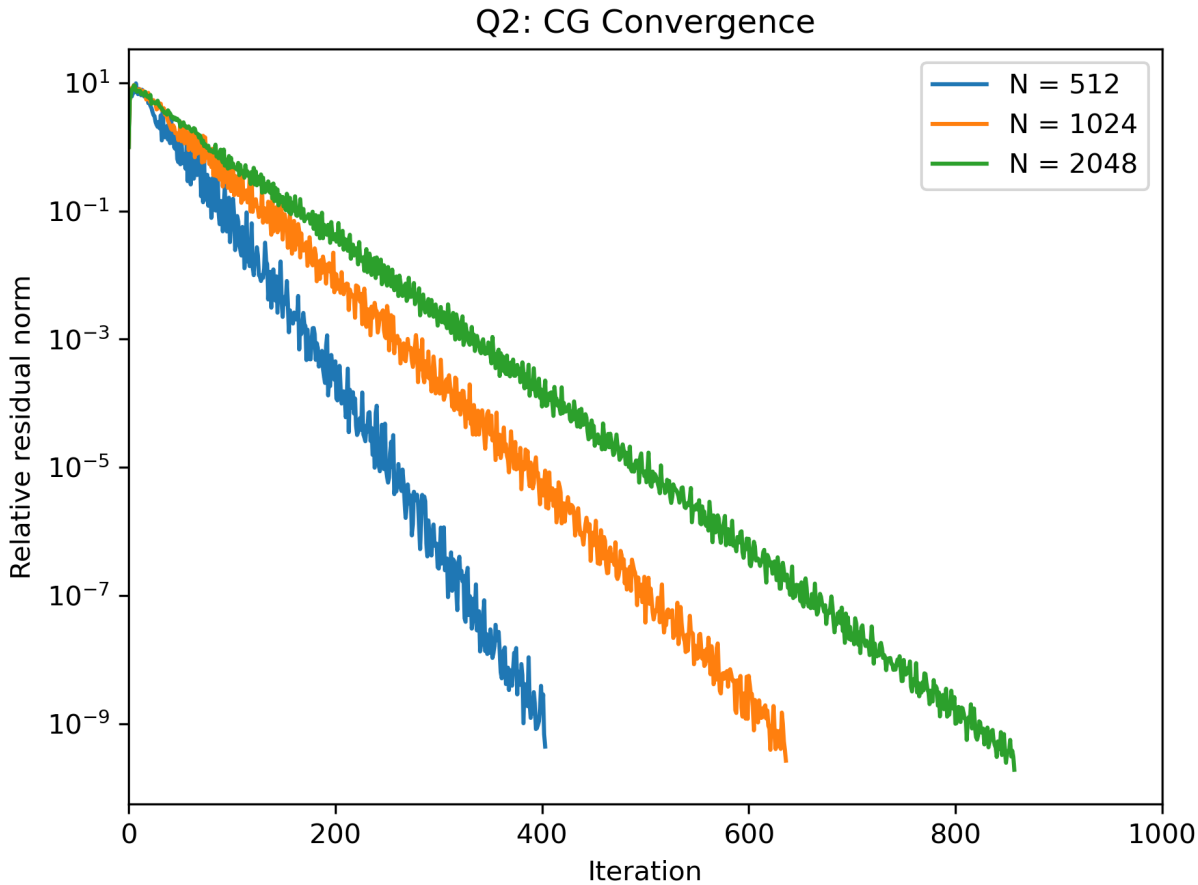

Figure 1: Relative residual norm as a function of number of CG iterations.

Question 3

The printed output from the Notebook shows that as the noise parameter decreases, the condition number A of increases. This is likely because reducing the noise of A causes the eigenvalue spectrum to spread over a wider interval. CG is a Krylov subspace method that at each iteration constructs a polynomial of increasing degree that must be small at all eigenvalues of A simultaneously. When the eigenvalues are widely spread, higher-degree polynomials are needed to achieve this, requiring more iterations to converge. Each order of magnitude decrease in noise increases the condition number by about a factor of 10, and approximately doubles the number of iterations needed.

Question 4

Figure 2 shows how the matrix A from Q4 converges much slower then its preconditioned version A_{new}.

Figure 3 shows how the eigenvalues of A are spread over a wider range, with most clustered near 10^{-1} and suddenly increasing exponentially for higher indexes, resulting in a large condition number. Due to the Cholesky preconditioning, A_{new}'s eigenvalues are much more tightly clustered, reducing the condition number significantly. CG constructs polynomials that must be small at all eigenvalues, so a tighter spectrum means a lower-degree polynomial suffices and convergence is faster. For this reason, A_{new} converges much faster than A.

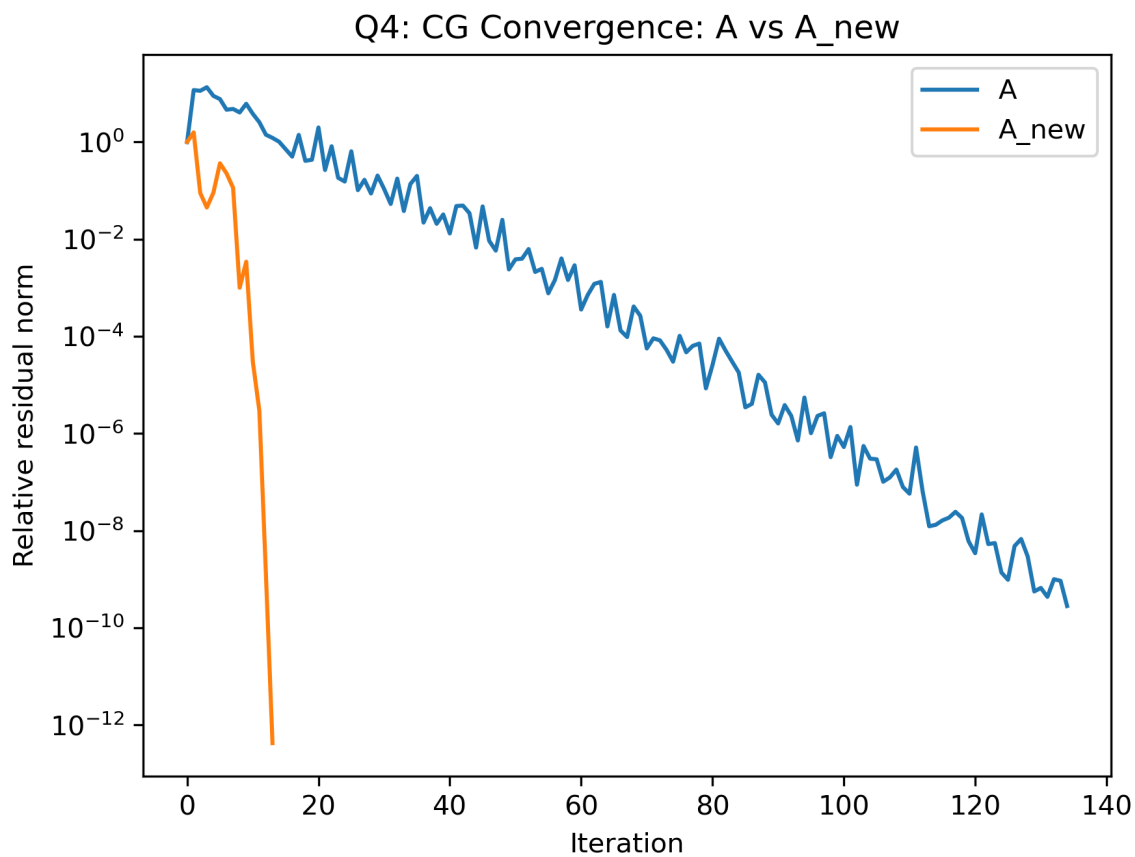

Figure 2: Relative Residual norm as a function of the number of CG iterations for Q4 matrix A (blue) and the preconditioned matrix A_{new} (orange).

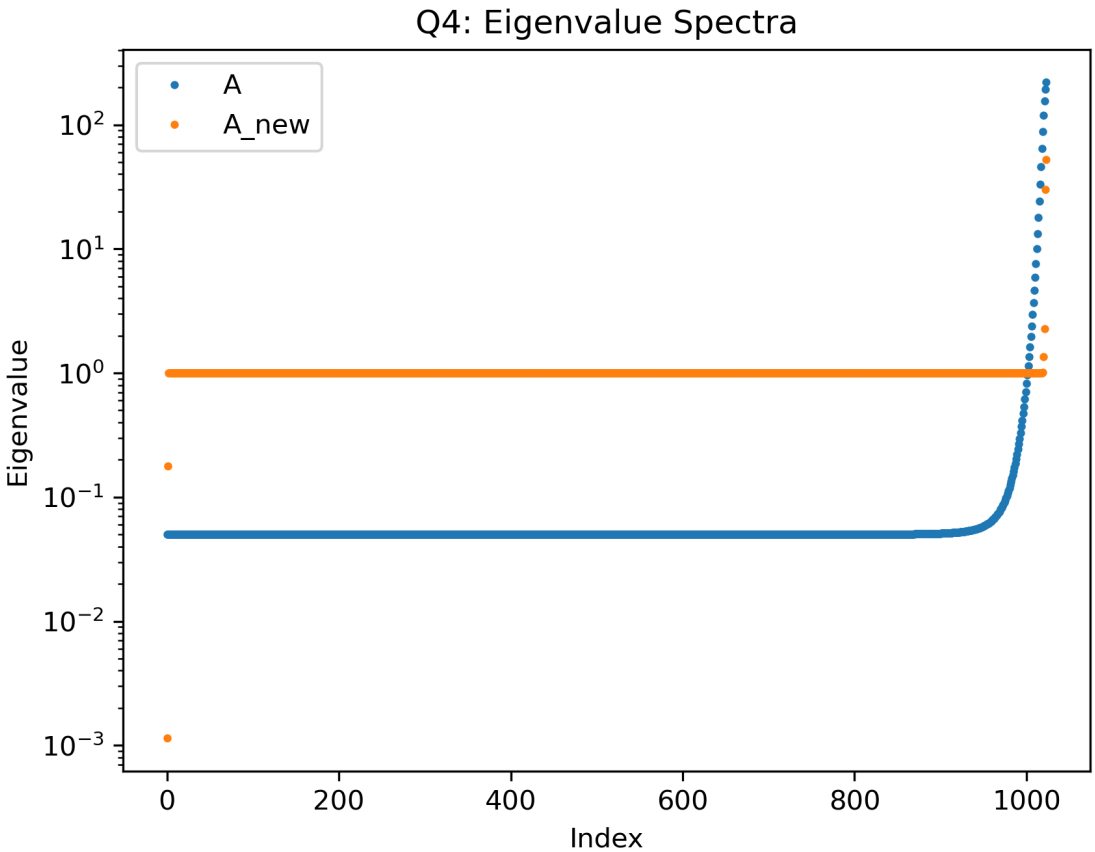

Figure 3: Eigenvalue spectra of matrix A (blue) and the preconditioned matrix A_{new} (orange).